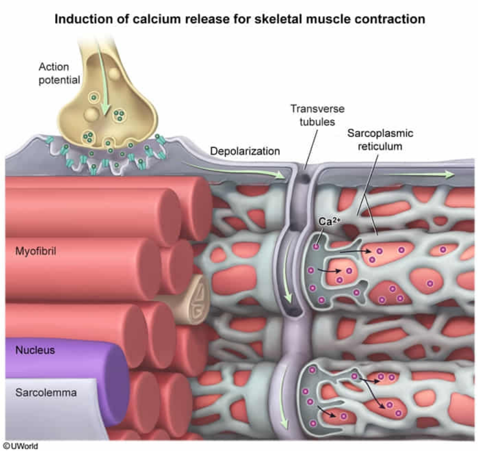
Which statement most accurately describes the role of T tubules in skeletal muscle cells?

- A. T tubules bind acetylcholine at the neuromuscular junction to generate a depolarizing stimulus. [9%]
- B. Depolarizing current reaches the sarcoplasmic reticulum by traveling down T tubules. [57%]
- C. Muscle contraction is driven by the sliding of T tubules across one another in the sarcomere. [20%]
- D. T tubules sequester Ca^{2+} out of the cytosol to prevent prolonged muscle contraction. [12%]

Correct
57% answered correctly

Explanation:



When acetylcholine (ACh) is released by the motor neuron at the neuromuscular junction, the following occur:

1. ACh binds and opens ligand-gated ion channels in the sarcolemma (the plasma membrane of the muscle cell) **(Choice A)**.

2. Na^{+} flows down its electrochemical gradient and into the cell through the channel, resulting in depolarization of the sarcolemma and generation of an action potential that propagates along the muscle fiber in all directions.

3. At certain locations along the muscle fiber, the sarcolemma burrows deep into the cells, forming a channel known as the **transverse (T) tubule**, which brings depolarizing current close to the sarcoplasmic reticulum (SR) **(Choice B)**. The SR is a specialized smooth endoplasmic reticulum responsible for regulating cytosolic Ca^{2+} levels within the muscle cell.

4. Action potential propagation through the T tubule ultimately leads to the opening of Ca^{2+} channels in the SR membrane. Because Ca^{2+} is more highly concentrated inside the SR than in the cytosol, the opening of these channels results in Ca^{2+} flowing down its concentration gradient and into the cytosol.

5. Cytosolic Ca^{2+} ions then bind to troponin, which allows the **actin and myosin** filaments of the sarcomere to **slide** across one another. The sliding of the filaments results in shortening of the sarcomere and overall muscle contraction **(Choice C)**.

6. The Ca^{2+} channels in the SR membrane close when the depolarizing stimulus ceases. **Active transport Ca^{2+} pumps** sequester the Ca^{2+} back into the SR, which allows the muscle to return to its relaxed state as cytosolic Ca^{2+} concentration falls **(Choice D)**.

Educational objective:

For a skeletal muscle cell to contract, Ca^{2+} must be released into the cytosol from the sarcoplasmic reticulum (SR). Ca^{2+} release is induced when a depolarizing current (action potential) runs along the sarcolemma and travels down the T tubules. This current causes the nearby SR to open its Ca^{2+} channels, allowing Ca^{2+} ions to flow into the cytosol and induce the sarcomeric actin-myosin interactions required for muscle contraction.

Time spent:0 sec
Subject:
Biology

QID:401498
System:
Musculoskeletal System